

lab05

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```
library(tidyverse)
```

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-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
```

```
# Obs. 3, 4
analisa_aeronave <- function(tail_number, arquivo) {

  # Obs. 6
  aeroportos <- read_csv("airports.csv", show_col_types = FALSE)

  # Obs. 5
  filtra_voos_do_aviao <- function(bloco, pos) {
    bloco %>% filter(TAIL_NUMBER == tail_number)
  }

  # Obs. 2, 5
  voos_do_aviao <- read_csv_chunked(
    arquivo,
    DataFrameCallback$new(filtra_voos_do_aviao),
    chunk_size = 50000,
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    show_col_types = FALSE
  )

# Parte 1.b
voos_ordenados <- voos_do_aviao %>%
  arrange(YEAR, MONTH, DAY, SCHEDULED_DEPARTURE) %>%
  mutate(velocidade_media = DISTANCE / (AIR_TIME / 60)) # Parte 1.d

# Obs. 6
trajeto_completo <- voos_ordenados %>%
  left_join(aeroportos, by = c("ORIGIN_AIRPORT" = "IATA_CODE")) %>%
  rename(latitude_origem = LATITUDE, longitude_origem = LONGITUDE) %>%
  left_join(aeroportos, by = c("DESTINATION_AIRPORT" = "IATA_CODE")) %>%
  rename(latitude_destino = LATITUDE, longitude_destino = LONGITUDE)

# Parte 1.c, Parte 1.d
mapa_base <- map_data("state")

grafico_trajeto <- ggplot() +
  geom_polygon(
    data = mapa_base,
    aes(x = long, y = lat, group = group),
    fill = "gray95", color = "white"
  ) +
  geom_segment(
    data = trajeto_completo,
    aes(
      x = longitude_origem, y = latitude_origem,
      xend = longitude_destino, yend = latitude_destino,
      linewidth = velocidade_media
    ),
    color = "steelblue", alpha = 0.7
  ) +
  geom_point(
    data = trajeto_completo,
    aes(x = longitude_origem, y = latitude_origem),
    color = "darkred", size = 2
  ) +
  coord_fixed(1.3) +
  labs(
    title = paste("Trajeto da aeronave", tail_number),
    linewidth = "Velocidade média (mph)"
  )

```

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) +
  theme_minimal()

# Obs. 7
list(tabela = trajeto_completo, grafico = grafico_trajeto)
}

resultado <- analisa_aeronave("N431WN", "flights.csv")

resultado$tabela

```

```

# A tibble: 1,938 x 44
  YEAR MONTH DAY DAY_OF_WEEK AIRLINE FLIGHT_NUMBER TAIL_NUMBER
  <dbl> <dbl> <dbl>      <dbl> <chr>          <dbl> <chr>
1  2015     1     1          4 WN             1762 N431WN
2  2015     1     1          4 WN             3564 N431WN
3  2015     1     1          4 WN             3564 N431WN
4  2015     1     1          4 WN             3564 N431WN
5  2015     1     2          5 WN             1793 N431WN
6  2015     1     2          5 WN             1793 N431WN
7  2015     1     2          5 WN             1793 N431WN
8  2015     1     2          5 WN             1793 N431WN
9  2015     1     2          5 WN             1793 N431WN
10 2015     1     2          5 WN             3144 N431WN
# i 1,928 more rows
# i 37 more variables: ORIGIN_AIRPORT <chr>, DESTINATION_AIRPORT <chr>,
#   SCHEDULED_DEPARTURE <chr>, DEPARTURE_TIME <chr>, DEPARTURE_DELAY <dbl>,
#   TAXI_OUT <dbl>, WHEELS_OFF <chr>, SCHEDULED_TIME <dbl>, ELAPSED_TIME <dbl>,
#   AIR_TIME <dbl>, DISTANCE <dbl>, WHEELS_ON <chr>, TAXI_IN <dbl>,
#   SCHEDULED_ARRIVAL <chr>, ARRIVAL_TIME <chr>, ARRIVAL_DELAY <dbl>,
#   DIVERTED <dbl>, CANCELLED <dbl>, CANCELLATION_REASON <chr>, ...

```

```
resultado$grafico
```

Warning: Removed 172 rows containing missing values or values outside the scale range (``geom_segment()``).

Warning: Removed 2 rows containing missing values or values outside the scale range (``geom_segment()``).

Warning: Removed 156 rows containing missing values or values outside the scale range (`geom_point()`).

